॥वसुधैव कुटुम्बकम्॥

# SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

**Symbiosis International (Deemed University)**

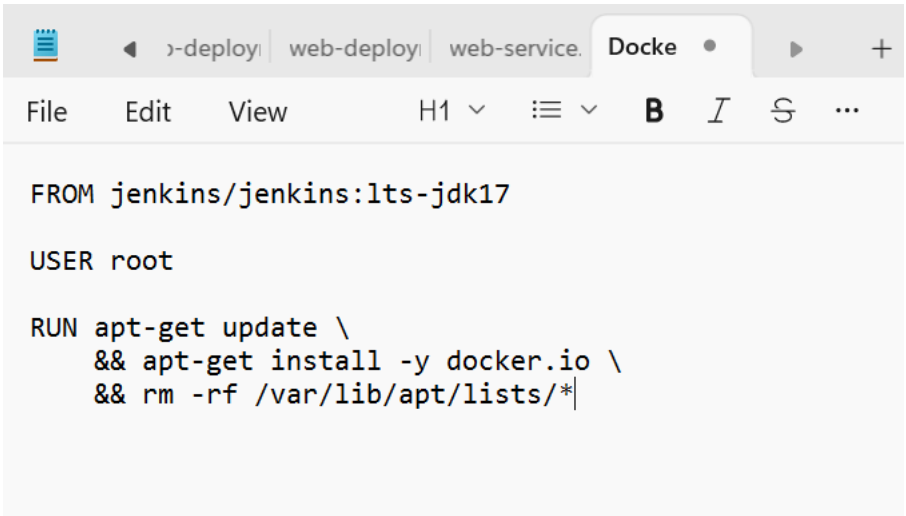
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

## Assignment No. 14

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|---|---|
| **Subject:** | DevOps Lab |
| **TCODE:** | TE7950 |
| **Name of Student** | **Yash Patel** |
| **PRN No.** | **23070122235** |
| **Branch** | CSE, Batch (2023-27), L4 |
| **Academic Year & Semester** | 25 – 26, Sem VII |
| **Date of Performance** | 23rd July, 2026 |
| **Title of Assignment:** | CI/CD: Jenkins Pipeline building Docker Image |
| **Theory:** | A Jenkins Pipeline automates the Continuous Integration and Continuous Deployment (CI/CD) process by executing a series of predefined stages such as source code checkout, build, test, and deployment. In this experiment, a Docker image was built automatically using Jenkins after fetching the application source code from a GitHub repository. The docker build command created a versioned Docker image using the Jenkins build number as the image tag, ensuring unique image versions for every pipeline execution. This integration of Jenkins and Docker enables consistent, repeatable, and automated application builds, reducing manual effort and improving software delivery efficiency.<br><br>**Commands Used**<br>• docker build -t my-jenkins-docker .<br>• docker run -d --name myjenkins --restart unless-stopped -u root -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home -v /var/run/docker.sock:/var/run/docker.sock my-jenkins-docker<br>• docker exec -it myjenkins docker --version<br>• docker exec -it myjenkins rm -rf /var/jenkins_home/workspace/Declarative-Pipeline-Demo<br>• git add .<br>• git commit -m "Add Dockerfile for Jenkins pipeline"<br>• git push origin main<br>• docker images |

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|---|---|
| **Implementation:** | **Step 1: Access and Edit Your Pipeline Configuration**<br><br>Accessed Jenkins Pipeline Configuration — Opened the existing Declarative Jenkins Pipeline and navigated to the Pipeline configuration section to modify the pipeline script.<br><br>**Step 2: Add the Build Docker Image Stage**<br><br>Added Docker Build Stage — Modified the existing Declarative Jenkins Pipeline by adding a new stage to build the Docker image using the docker build command with the Jenkins build number as the image tag.<br><br>**Step 3: Run the Pipeline**<br><br>Executed Jenkins Pipeline — Successfully executed the Jenkins Declarative Pipeline containing the Docker image build stage. The pipeline completed successfully, and the Docker image was built and tagged without errors.<br><br>**Step 4: Verify the Docker Image**<br><br>  <br><br><br>  <br> |

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>type Dockerfile
FROM jenkins/jenkins:lts-jdk17

USER root

RUN apt-get update ^
    && apt-get install -y docker.io ^
    && rm -rf /var/lib/apt/lists/*
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>notepad Dockerfile

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>type Dockerfile
FROM jenkins/jenkins:lts-jdk17

USER root

RUN apt-get update && apt-get install -y docker.io && rm -rf /var/lib/apt/lists/*
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>docker build -t my-jenkins-docker .
[+] Building 70.0s (6/6) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                0.0s
 => => transferring dockerfile: 167B                                                                0.0s
 => [internal] load metadata for docker.io/jenkins/jenkins:lts-jdk17                                0.1s
 => [internal] load .dockerignore                                                                   0.0s
 => => transferring context: 2B                                                                     0.0s
 => [1/2] FROM docker.io/jenkins/jenkins:lts-jdk17@sha256:cce1eeb79902722f3cf7a8831c41e95c39f4a42b9a2981959630cc6  0.3s
 => => resolve docker.io/jenkins/jenkins:lts-jdk17@sha256:cce1eeb79902722f3cf7a8831c41e95c39f4a42b9a2981959630cc6  0.0s
 => [2/2] RUN apt-get update && apt-get install -y docker.io && rm -rf /var/lib/apt/lists/*        49.2s
 => exporting to image                                                                             20.1s
 => => exporting layers                                                                            17.0s
 => => exporting manifest sha256:597410d390ef48dfa3626ccd4213bf8b63462b7d5f8f13d79f51ff5f18fe6488   0.0s
 => => exporting config sha256:80be63e50530ea401f93c966aef749b8425d73762f97c1cb7e7944ab0f2d8e4e     0.0s
 => => exporting attestation manifest sha256:1d34aa856511e12ff3943003908dc47050fa19ce36342aea7b1f0303331c3250  0.0s
 => => exporting manifest list sha256:91ff8c2acbaa986b2dbc04b2768af8acfaa24f7bdd2f6def5ad20e72deff980b  0.0s
 => => naming to docker.io/library/my-jenkins-docker:latest                                         0.0s
 => => unpacking to docker.io/library/my-jenkins-docker:latest                                      2.8s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/sehukspcnxnjjocqhiygavge9

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>
```

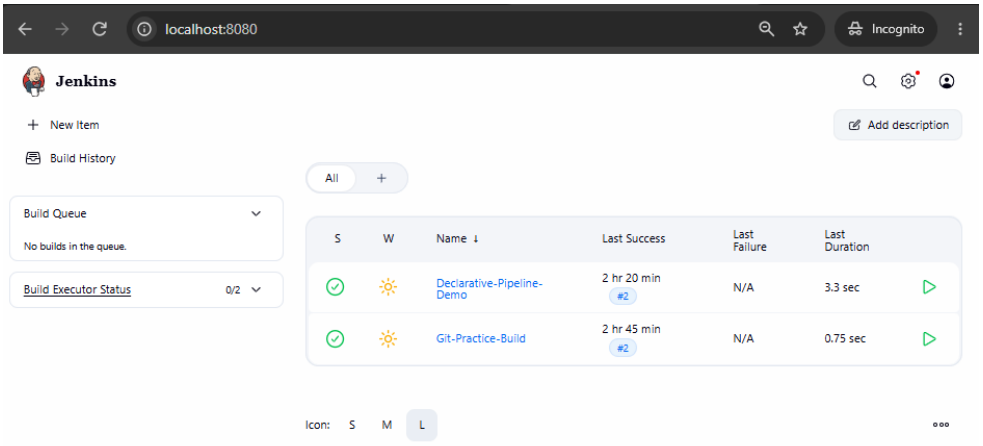
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>docker images my-jenkins-docker
                                                                                  i Info →  U In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
my-jenkins-docker:latest       91ff8c2acbaa       1.36GB          425MB

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS          PORTS
                                                                  NAMES
857721e057d1   my-jenkins-docker   "/usr/bin/tini -- /u…"   13 seconds ago   Up 12 seconds   0.0.0.0:8080->8080/tcp, [::
]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
8f7a2145adab   my-flask-app:v2    "python app.py"          59 minutes ago   Up 59 minutes   0.0.0.0:8082->5001/tcp, [::
]:8082->5001/tcp                                                  flask-volume-demo2
9cfdd009c213   my-flask-app       "python app.py"          2 hours ago      Up 2 hours      0.0.0.0:5000->5000/tcp, [::
]:5000->5000/tcp                                                  my-flask-app-container
6469f5fb413a   postgres:15        "docker-entrypoint.s…"   2 months ago     Up 3 hours      0.0.0.0:5432->5432/tcp, [::
]:5432->5432/tcp                                                  flaskapp-db-1

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>docker exec -it myjenkins docker --version
Docker version 26.1.5+dfsg1, build a72d7cd

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug myjenkins
    Learn more at https://docs.docker.com/go/debug-cli/
```



Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```groovy
pipeline {
    agent any

    parameters {
        string(
            name: 'GREETING_NAME',
            defaultValue: 'World',
            description: 'The name to use in the greeting'
        )
    }

    stages {
        stage('Greeting') {
            steps {
                echo "Hello ${params.GREETING_NAME}"
```

Use Groovy Sandbox ?

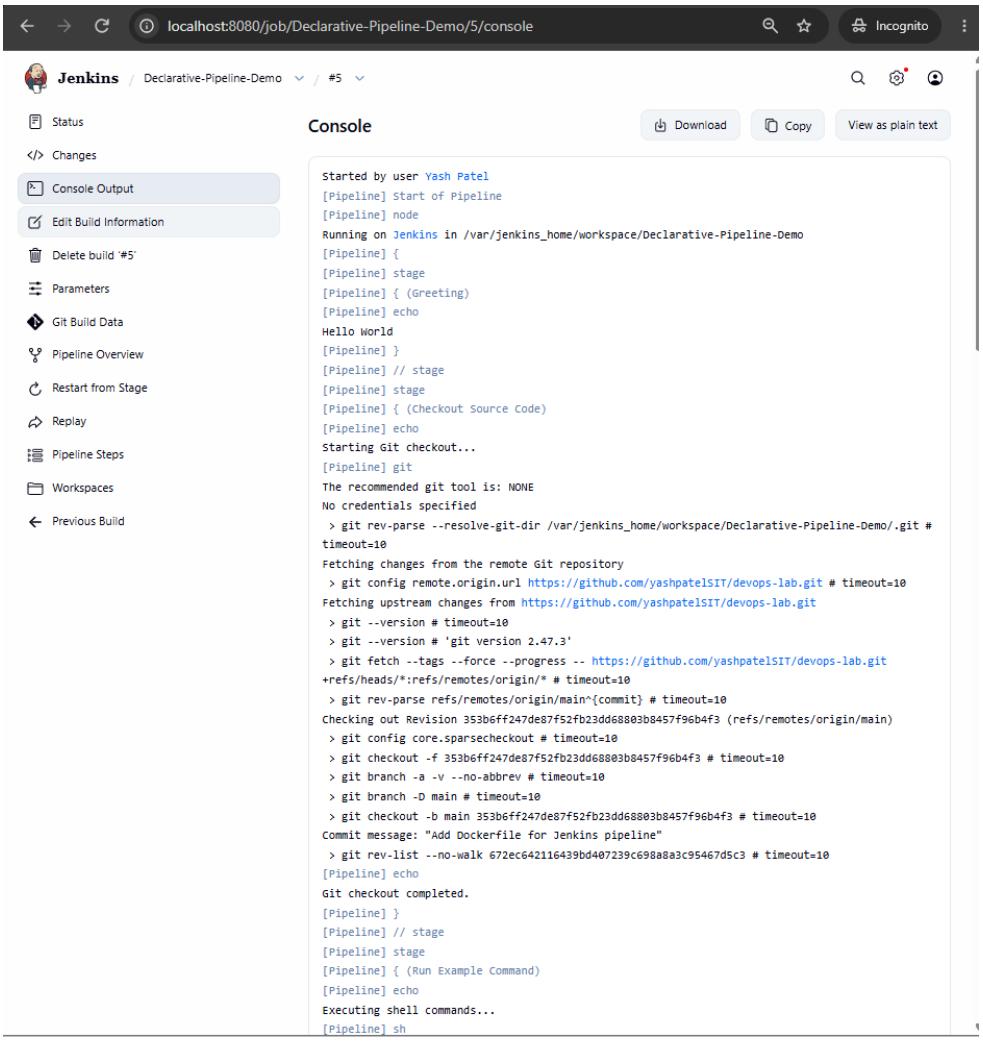
Pipeline Syntax

```groovy
        stage('Build Docker Image') {
            steps {
                sh 'docker build -t my-flask-app:${BUILD_NUMBER} .'
            }
        }
    }

    post {
        always {
            echo 'Pipeline finished.'
        }
    }
}
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Dockerfile
        new file:   app.py
        new file:   requirements.txt


C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 14>
```

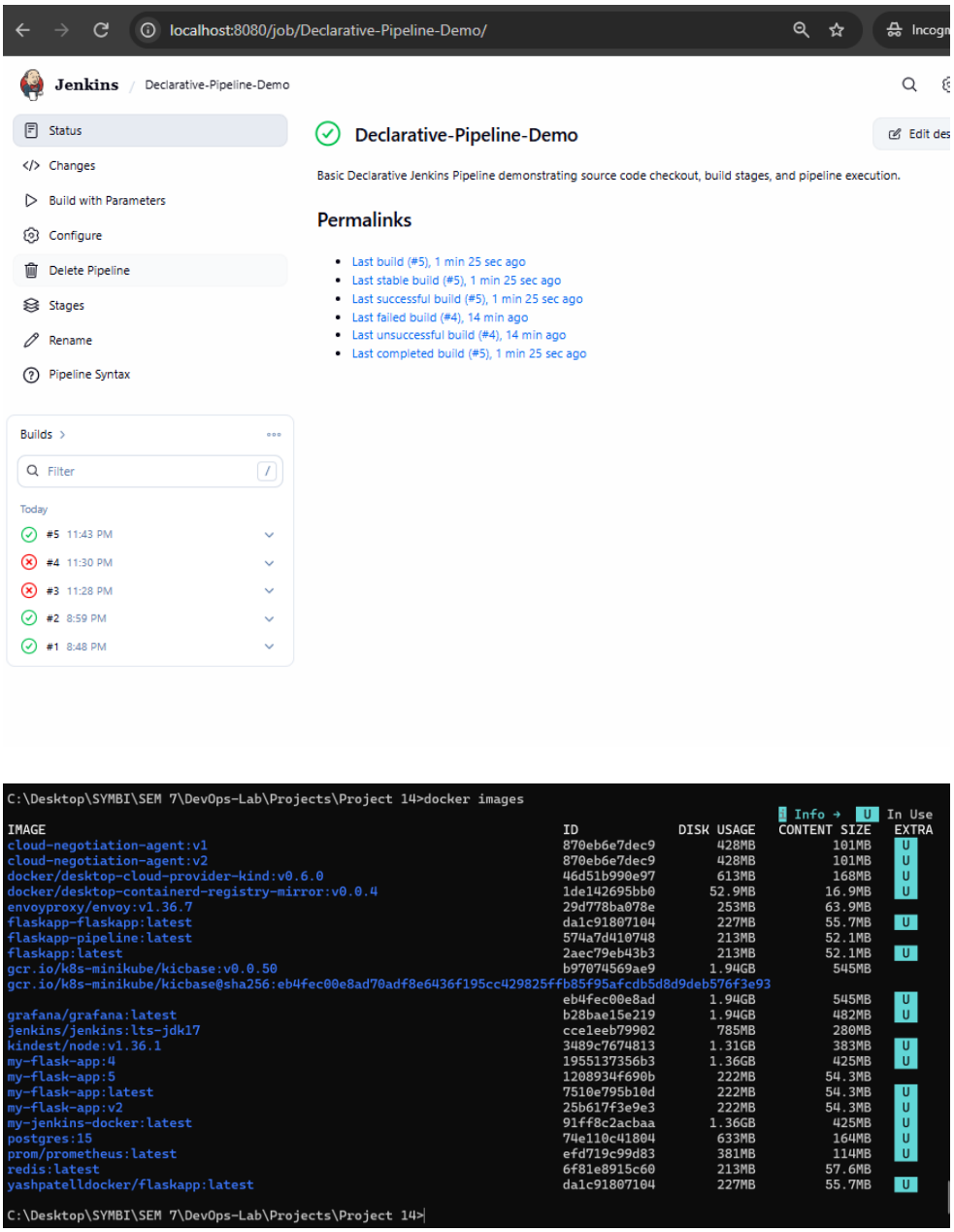



```
#8 6.615 [notice] To update, run: pip install --upgrade pip
#8 DONE 6.8s

#9 [5/5] COPY app.py .
#9 DONE 0.1s

#10 exporting to image
#10 exporting layers
#10 exporting layers 0.8s done
#10 exporting manifest sha256:29627f1d5be09f57a6d03502ce8ce5ecafdb4e8b6c5163a223e2bc3bcb22f3b3
0.0s done
#10 exporting config sha256:c526bd5f5704adfbe3dcd2c22ec5cee340d239cf6616f4ead9f0f4cd2fe18874
0.0s done
#10 exporting attestation manifest
sha256:66b0458f90e37251f6a9f92a60fb37bb094be3538c7bdbbb78d91b004623b22b 0.0s done
#10 exporting manifest list
sha256:1208934f690b755bfb93b6f7f976054e15be0f300eb7c7acc914dd974bd99d82 0.0s done
#10 naming to docker.io/library/my-flask-app:5
#10 naming to docker.io/library/my-flask-app:5 done
#10 unpacking to docker.io/library/my-flask-app:5
#10 unpacking to docker.io/library/my-flask-app:5 0.3s done
#10 DONE 1.3s
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline finished.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.3




Subject Faculty Name: Prof. Arijit Dutta